

Mining Insights from Customer Support Tickets: A Comparative Text Analytics Pipeline

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Abstract—XXX
Index Terms—XXX

I. INTRODUCTION

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II. EXPLORATORY DATA ANALYSIS

This section presents a systematic EDA of the customer support ticket dataset, covering data quality, text length distribution, word frequency patterns, class distributions, and customer satisfaction characteristics. All results are based on 8,077 records retained after deduplication of the original 8,469-row dataset.

A. Dataset Overview and Preprocessing

The raw dataset contains 8,469 rows and 17 columns. For this study, four fields were retained: Ticket Description (free-text input, renamed text), Ticket Type (five-class label), Customer Satisfaction Rating (1–5 ordinal score), and Ticket Priority (four-class label).

The CSV file used latin-1 encoding (ISO-8859-1). Loading with the default UTF-8 codec produced a UnicodeDecodeError at byte position 26191, which was resolved by specifying encoding='latin-1'.

After removing null descriptions and deduplicating on Ticket Description, **8,077 records** were retained a 4.6% reduction.

The following preprocessing steps were applied:

- 1) **Lowercasing**: all text converted to lowercase.
- 2) **Template placeholder removal**: the regex pattern `\{[^\}]+\}` stripped unfilled variables such as `{product_purchased}`, which would otherwise dominate token frequencies.
- 3) **Special character removal**: non-alphabetic characters removed, with hyphens preserved for compound terms (e.g. *follow-up*).
- 4) **Whitespace normalisation**: multiple spaces collapsed.

A custom stopwords list extended scikit-learn's ENGLISH_STOP_WORDS with domain-specific but semantically empty tokens: `{product_purchased, im, ive, assist, having, product}`. Numeric tokens were excluded as they predominantly represent billing codes and order numbers with no value for issue categorisation.

B. Text Length Distribution

Table I presents descriptive statistics for the cleaned text length in words. The mean is 44.76 words (std = 8.14), with an interquartile range of only 10 words (Q1 = 41, Q3 = 51). As shown in Fig. 1, the distribution is unimodal and approximately normal, centred around 47–50 words. This narrow spread is atypical of genuine user-generated text, where lengths typically vary by orders of magnitude, and strongly suggests template-driven generation. The constrained length also means that document truncation or padding is unnecessary for any downstream representation method.

TABLE I
DESCRIPTIVE STATISTICS: CLEANED TEXT LENGTH (WORDS)

Statistic	Value
Count	8,077
Mean	44.76
Std Dev	8.14
Min	8
25th pct.	41
Median	47
75th pct.	51
Max	60

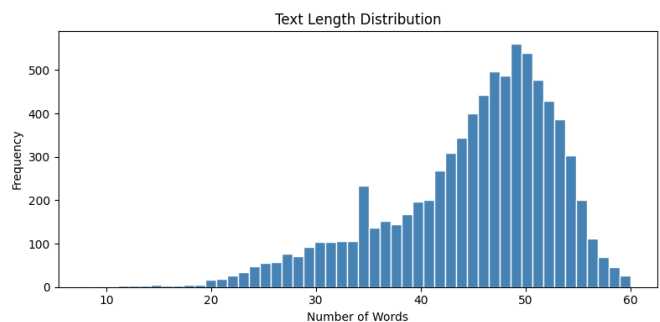


Fig. 1. Distribution of cleaned ticket description lengths ($n = 8,077$). The unimodal shape with low variance is consistent with template-generated text rather than natural user writing.

C. Word Frequency Analysis

1) *Top-20 Unigrams*: Table II lists the 20 most frequent tokens after applying the custom stopwords list. The term *issue* (10,947 occurrences) dominates, appearing nearly five

times more frequently than the second-ranked token *problem* (2,213), and is present in virtually every ticket. This extreme skew indicates that *issue* functions as a universal template phrase rather than a discriminative signal. The remaining meaningful tokens—*software*, *account*, *data*, *device*, *update*—suggest technical and account-management problems as the primary customer issue themes.

TABLE II
TOP-20 UNIGRAMS AFTER STOPWORD AND TEMPLATE REMOVAL

Rank	Token	Frequency
1	issue	10,947
2	problem	2,213
3	data	1,605
4	software	1,476
5	account	1,447
6	steps	1,340
7	help	1,152
8	persists	1,132
9	resolve	1,130
10	noticed	1,123
11	update	1,109
12	device	1,060
13	unable	1,044
14	need	993
15	using	982
16	tried	968
17	use	941
18	started	935
19	times	928
20	support	903

2) *Top-20 Bigrams*: Table III reports the top-20 bigrams extracted using CountVectorizer with `ngram_range=(2, 2)` and English stopwords applied. Two structural patterns stand out. First, *having issue* (5,790) and *issue assist* (5,662) appear in the majority of tickets, confirming a shared template opening. Second, ranks 16–20 all share an identical frequency of 478, indicating that the phrase “*facing intermittent ... acts unexpectedly*” is copied verbatim into exactly 478 tickets. Both patterns further confirm the synthetic, template-driven nature of the corpus.

D. Ticket Type, Priority, and Cross-Type Vocabulary

1) *Class Distributions*: Table IV reports the distribution across Ticket Type and Ticket Priority. Both are near-perfectly uniform: the five ticket types each account for approximately 20% of records (imbalance ratio 1.06:1), and the four priority levels each account for approximately 25% (ratio 1.07:1). In a real-world corpus, significant skew would be expected—for example, technical issues typically outnumber refund requests. This uniformity confirms balanced synthetic class sampling, and means that $K = 5$ is a well-motivated initialisation for K-Means clustering, with the ground-truth labels serving as a natural external validation benchmark.

2) *Cross-Type Vocabulary Overlap*: To assess whether ticket descriptions are linguistically distinguishable by type, the top-10 unigrams were extracted independently for each ticket type. As shown in Table V, the top-5 tokens are virtually identical across all five categories, with only minor rank

TABLE III
TOP-20 BIGRAMS (ENGLISH STOP WORDS REMOVED)

Rank	Bigram	Frequency
1	having issue	5,790
2	issue assist	5,662
3	issue persists	1,130
4	ive noticed	1,105
5	ive tried	899
6	resolve problem	712
7	ive checked	689
8	troubleshooting steps	655
9	ive recently	634
10	works fine	486
11	soon possible	486
12	problem didnt	480
13	hoping resolve	479
14	didnt help	479
15	issue facing	478
16	facing intermittent	478
17	intermittent works	478
18	fine times	478
19	times acts	478
20	acts unexpectedly	478

TABLE IV
TICKET TYPE AND PRIORITY CLASS DISTRIBUTIONS

Category	Count	Share (%)
<i>Ticket Type</i>		
Refund request	1,659	20.5
Technical issue	1,651	20.4
Cancellation request	1,619	20.0
Product inquiry	1,577	19.5
Billing inquiry	1,571	19.5
<i>Ticket Priority</i>		
Medium	2,090	25.9
Critical	2,030	25.1
High	2,001	24.8
Low	1,956	24.2

differences at positions 3–5. This suggests that bag-of-words representations such as TF-IDF are unlikely to produce well-separated clusters on this corpus.

TABLE V
TOP-5 UNIGRAMS PER TICKET TYPE (CUSTOM STOPWORDS REMOVED)

Rank	Tech.	Billing	Cancel.	Product	Refund
1	issue	issue	issue	issue	issue
2	problem	problem	problem	problem	problem
3	steps	data	data	data	data
4	data	account	account	software	software
5	software	software	software	account	account

E. Customer Satisfaction Rating Analysis

Table VI summarises satisfaction ratings by ticket type. Two issues stand out. First, 67.0% of records lack a rating, with only 2,662 of 8,077 records containing valid scores. Second, Fig. 2 shows that the valid ratings are distributed almost uniformly across values 1–5 (each appearing approximately 515–555 times), and mean ratings cluster tightly between 2.94

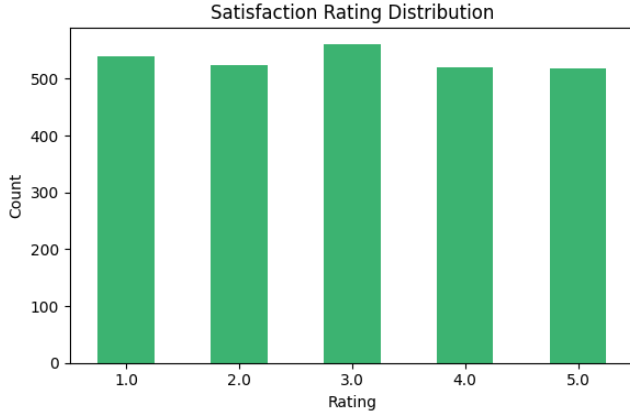


Fig. 2. Customer Satisfaction Rating frequency distribution (ratings 1–5). The near-uniform spread across all five values is inconsistent with genuine user feedback, indicating random assignment.

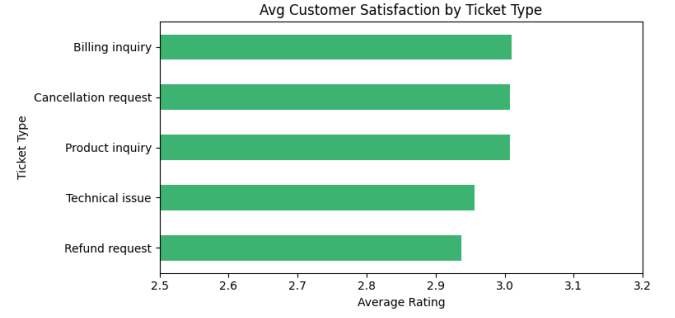


Fig. 3. Average Customer Satisfaction Rating by Ticket Type (x-axis range 2.5–3.2). All five categories achieve virtually identical mean scores (≈ 3.0), confirming the absence of meaningful inter-group signal.

and 3.01 across all ticket types with a standard deviation of approximately 1.41. This pattern is consistent with randomly assigned values rather than genuine user sentiment. Consequently, as further illustrated in Fig. 3, there are no meaningful differences in average satisfaction across ticket categories, rendering this field unsuitable as a predictive target or analytical dimension.

TABLE VI
CUSTOMER SATISFACTION RATING BY TICKET TYPE

Ticket Type	Mean	Count	Std Dev
Billing inquiry	3.01	522	1.41
Cancellation request	3.01	497	1.43
Product inquiry	3.01	513	1.37
Technical issue	2.96	557	1.41
Refund request	2.94	573	1.43
Overall missing rate	67.0%		

F. Hypotheses for Method Comparison

The EDA findings motivate two hypotheses that will guide the comparative evaluation in subsequent sections.

H1 (Text Representation): Given the near-total vocabulary overlap across ticket types (Section II D), TF-IDF vectors will produce poorly separated clusters because term frequency alone carries minimal inter-class discriminative signal. Sentence-level embeddings, which encode contextual and semantic relationships beyond surface token frequencies, are expected to yield more coherent and interpretable cluster assignments.

H2 (Issue Discovery Method): With five balanced ticket-type classes, K-Means clustering ($K = 5$) is expected to achieve reasonable alignment with ground-truth labels when applied to embedding representations. Topic modelling (LDA/BERTopic) is hypothesised to produce finer-grained, more interpretable issue summaries by capturing latent thematic structure within and across ticket categories, even if its external label agreement is lower than clustering.

III. TEXT PREPROCESSING AND REPRESENTATION

A. TF-IDF Representation

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B. Sentence Embedding Representation

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C. Comparison and Discussion

XXX

IV. ISSUE DISCOVERY METHODS

A. Clustering

1) *K-Means*: XXX

2) *Hierarchical Agglomerative Clustering (HAC)*: XXX

B. Topic Modelling

1) *Latent Dirichlet Allocation (LDA)*: XXX

2) *BERTopic*: XXX

V. RESULTS AND ACTIONABLE INSIGHTS

A. Main Issue Categories Identified

XXX

B. Actionable Insights for Product and Service Improvement

XXX

VI. EVALUATION AND DISCUSSION

A. Consistency and Robustness

XXX

B. Coherence and Interpretability

XXX

C. Alignment with Ground Truth and EDA Findings

XXX

D. Impact of Design Choices

XXX

VII. CONCLUSION

REFERENCES

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